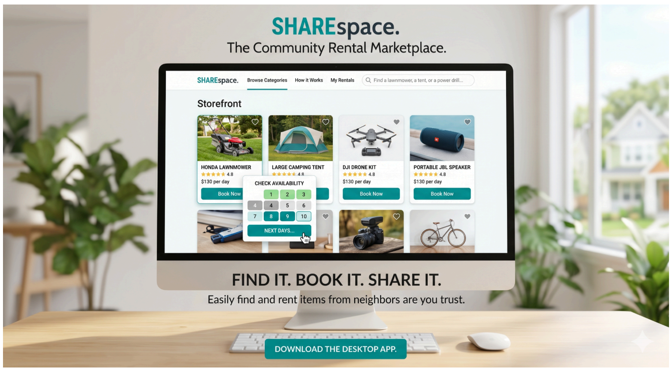
Project 9: ShareSpace
Peer-to-Peer Rental Marketplace

1. Scenario



The sharing economy promotes the efficient use of resources by allowing individuals to rent out items that are not in constant use. ShareSpace is a community-driven marketplace platform designed to facilitate these transactions within local neighborhoods. The objective is to develop a robust prototype that manages a diverse inventory of rentable goods—ranging from specialized tools to leisure equipment—while coordinating the logistical aspects of bookings, user accounts, and financial calculations to ensure a reliable exchange between lenders and renters.

2. Functional Objectives

The application provides the following capabilities:

- **Asset Listing Management:**
  Users register items for rent, capturing detailed descriptions, usage conditions, and daily rental rates.

- **User Profile Administration:**
  The system manages distinct accounts for lenders and renters, tracking their participation history and current marketplace status.

- **Booking and Availability Coordination:**
  A centralized scheduling system allows users to check the real-time availability of items and reserve them for specific temporal windows.

- **Rental Lifecycle Tracking:**
  The application maintains a comprehensive record of active and historical transactions, including the calculation of total costs based on rental duration.

- **Data Persistence:**
  All item listings, user account data, and historical booking records are stored permanently. The system ensures that the marketplace environment remains consistent and restorable across application sessions.

3. Engineering Requirements

The following technical objectives must be met through independent architectural decisions:

- **Rentable Asset Modeling:**
The team must design a class hierarchy that represents the diverse nature of items in the marketplace. This includes modeling specific requirements for different categories—such as maintenance logs for mechanical tools or battery health for electronics—while maintaining a unified interface for the rental process.

- **Architectural Separation:**
The implementation requires a strict boundary between the marketplace data models, the graphical user interface, and the transaction logic. This ensures that the core rules governing rentals and availability remain independent of the visual layout.

- **Transactional Integrity and Validation:**
The system must implement rigorous validation logic to prevent scheduling conflicts, such as overlapping bookings for the same item. A clear strategy for handling financial calculations and invalid date ranges is required to maintain the reliability of the marketplace.

- **Logic Verification:**
Automated tests are required to verify the accuracy of the rental cost engine and the integrity of the availability state transitions.

4. Trust Systems and Market Analytics

The prototype includes a layer for community reliability and platform insights:

- **Trust and Rating Logic:**
The system must implement a mechanism for users to provide feedback on completed transactions. This involves an analytical logic that aggregates individual ratings to calculate and display the average reputation of both items and users within the interface.

- **Documentation Automation:**
The application provides a summary generator that creates a formal rental record upon booking confirmation. This requires a logic capable of compiling item details, user information, and temporal data into a standardized, human-readable format.

- **Marketplace Visualization:**
A visual dashboard provides an overview of current marketplace activity. This includes using graphical components to display data such as item popularity or seasonal rental trends through interactive charts and color-coded availability calendars.